News Shocks in Prediction Markets

Real results, 2026-10-14

Avi Kabra

The question driving this thesis

Do news shocks about a company produce a measurable, predictable price reaction in that company's Polymarket contracts?

Everything that follows answers this question directly.

Context

- Data: 6,928 resolved contracts across 86 companies (Apr 2024-Aug 2026), matched against 546,657 GDELT articles
- No existing tool answers the question above directly, so this project builds two new methods
- GDELT only timestamps articles to the day, so every result below measures the 24-hour price reaction, the only window this feed supports

Method: the mechanics

Matching picks the right article for a contract, combining text similarity and trading-volume spikes:

match_quality = w1 · logit(cosine similarity) + w2 · z(volume spike), then sigmoid

Consistency forces a company's related contracts (above $150, $160, $170) to react in the same direction, weighted by volume, via isotonic regression (PAVA).

Method: why these, and why they should work

Both methods are new combinations of established tools, not inventions from scratch.

Combining independent signals in log-odds space is standard practice (the same math behind naive Bayes classifiers). Volume spikes as a marker of real news arriving is a standard idea in market microstructure research, going back to classic event-study methodology. Isotonic regression is a long-established tool for order-restricted inference (Barlow, Bartholomew, Bremner & Brunk, 1972), and the logic that related contracts should price consistently is the same logic behind no-arbitrage pricing.

What's new: applying both combinations to this specific problem, matching news to prediction-market contracts and enforcing no-arbitrage-style consistency across a strike ladder, which we are not aware of being done before.

Result 1: contracts do react to news, with one caveat found since

Method	Spearman ρ vs. actual price move	n
Matching score (text + volume)	**0.509**	14,717
Text similarity alone	0.233	14,717

This is half the answer to our question: a company's news and its market's trading behavior are measurably linked. Every later result depends on this step, it is how we built a trustworthy dataset of article-reaction pairs at all.

New finding (next section): part of the 0.509 number is inflated. Validated against the real stock's own reaction instead of Polymarket's own data, the matching score drops to ρ=0.029; text similarity alone holds up better (ρ=0.092). The volume component was likely riding a coincidence, not genuinely validating match quality.

Result 2: the consistency correction doesn't help, so we don't use it

Category	R^2, raw reaction	R^2, after correction
Price-target	0.084	0.073
Valuation	0.746	0.363

We expected this to help: related contracts should price consistently, by no-arbitrage logic. It likely hurts because the correction smooths away real, contract-specific variation the prediction models depend on. We predict from the raw reaction, not the corrected one.

Result 3: but the reaction isn't predictable in advance

Category	Best model	R^2	n (test)
All (pooled)	Linear	−0.010	5,151
Corporate event	Linear	−0.021	3,157
Price-target	LSTM	**+0.010**	598
Revenue-target	TCN	−0.016	9
Valuation	Linear only	−0.255	small

This is the other half of the answer. If Polymarket prices already reflect public news quickly, this is exactly the pattern we'd expect: a real reaction (Result 1), but no way to forecast its size from the article alone.

Result 4: one of four pre-registered predictions holds up

Prediction	Tests whether…	Result
Richer embeddings beat simpler ones	text representation matters	Not supported
Best model differs by category	architecture choice matters	Not supported
Quantitative articles beat high-attention ones	article type matters	Supported (n=28)
Best ladder category beats corporate events	ladder structure adds signal	Not supported

Tested with bootstrap resampling clustered by real-world event, so duplicate coverage of one story isn't counted as independent evidence.

A harder question: does it match the real world?

Every result so far compares Polymarket against **itself** — the matching score vs. Polymarket's own price move, the prediction models vs. Polymarket's own target. None of it checks whether the real, publicly-traded company also reacted, or whether Polymarket's reaction is proportionate.

Pulled real daily stock prices for the 78 public companies in our universe (yfinance) and re-ran both novel-math threads against this external benchmark.

Result 5: a real relationship, but not proportionate

Test	Spearman ρ	Significant?
Direction of Polymarket's move vs. the real stock's move	0.119	Yes (n=10,905)
Size of Polymarket's move vs. **size** of the real stock's move	0.015	No

Polymarket leans the right direction, weakly. But the size of its reaction tells you almost nothing about how big the real-world move was. A company-clustered bootstrap (correcting for shared companies/time windows across articles) confirms the directional result is real, not a statistical artifact: 95% CI [0.047, 0.188], excludes zero.

Result 6: a tradeable signal, isolated to one contract type

Twenty-three model specifications were tried against the full external benchmark before finding one that survives scrutiny (literature-standard forms, a novel match-quality moderator, window and liquidity variants, a digital-option-pricing structure). The signal turned out to be concentrated in **price-target contracts** specifically, where the strike is a real stock price in the same units as the real stock:

Check	Result
Per-split correlation (train/val/test, independent)	0.18 / 0.14 / 0.18, all significant
Company-clustered bootstrap (price-target only)	95% CI [0.043, 0.288], excludes zero
Backtest P&L vs. 5,000 random permutations	100th percentile (p=0.0002)
Bonferroni-corrected for all 23 forms tried	**p=0.0044, still significant**

Result 6: the equation, in named terms not bare decimals

`Δlogit(P) = κ · D · (ΔlnS / σ) + δ`

Symbol	Meaning	Value
D	theoretical slope a frictionless digital option would have (logistic–normal approximation; Birnbaum 1968)	**1.702** (external constant, not fit)
κ	efficiency ratio: how much of that theoretical sensitivity Polymarket actually realizes	**0.093**
δ	average 24h drift with no real-world move, consistent with favorite-longshot overpricing correction (Snowberg & Wolfers 2010)	**−0.184** (empirical)

Headline: these contracts move at ~9.3% of the rate a theoretically efficient digital

Result 6, continued: what this is and isn't

- **Is**: a real, cluster-robust, theoretically-motivated signal, backtested and statistically significant after correcting for every model tried
- **Isn't**: a precisely calibrated prediction model (R^2 stays negative — expected for an effect this size, not a failure) or proven beyond price-target contracts
- **Real limitation**: only 16 companies back this finding; held out exactly 4 companies never seen in training (bank stocks) and the signal held

So, does Polymarket respond to news shocks?

Yes, but more narrowly than it first looked. Polymarket's own data shows a real reaction (Result 1, with the volume-term caveat), and that reaction is not predictable in advance from the article alone (Results 2-4).

Checked against the real world, the picture sharpens further. The reaction tracks the real stock's direction, weakly but robustly (Result 5) — not its size. And a genuinely tradeable version of this signal exists, but only for price-target contracts specifically (Result 6), surviving the most adversarial testing we could throw at it.

Scoping decisions

- **market_cap_ladder** (4 markets): dropped, too small to mean anything
- **valuation_ladder** (110 markets): kept, flagged as below our 200-market floor, everything else about it clears comfortably
- **revenue_ladder**: given its own data split so it could be trained at all, rather than excluded

Limitations

- GDELT is free but noisy, and day-precision is a structural limit of this feed, not a gap to close
- `resolved_at` and `parent_event_id` are each about 40% backfilled from Polymarket's API; fallbacks cover the rest
- The revenue-target result rests on 19 training examples: real, but small
- The price-target trading signal (Result 6) rests on 16 companies total; transaction costs are literature-based, not Polymarket's measured spreads

Recommendation

- Lead the thesis with the matching result (Result 1) and the price-target trading signal (Result 6): the two clean, genuine, adversarially-tested contributions
- Present the consistency correction, the model comparison, and the external-benchmark proportionality check as honest negative/mixed results, not failures
- Keep the 200-market floor as the rule, with `valuation_ladder` as a documented, explicit exception
- Measuring Polymarket's actual historical bid-ask spread would replace Result 6's literature-based cost estimate with a real one — the highest-leverage next step